

Mithin Sagar S

Chennai, India | +91 63817 90265 | mithinsagar@gmail.com | github.com/mithinsagar

Education

Vellore Institute of Technology, Chennai

B.Tech, Computer Science and Engineering (Artificial Intelligence and Machine Learning)

Aug 2023 – May 2027

Chennai, India

Experience

Machine Learning Intern

Indira Gandhi Centre for Atomic Research (IGCAR)

May 2025 – Jun 2025

Kalpakkam, India

- Built an automated weld-inspection pipeline in Python chaining image preprocessing, model inference, and result generation into one reproducible workflow, replacing a manual process.
- Standardized data-handling and preprocessing routines for industrial image datasets, improving repeatability across batches and validating stable end-to-end runs on constrained inspection hardware.

Projects

AWS Cloud Resource Governance Platform | *Python, Boto3, AWS Lambda, EventBridge, SNS, Flask, REST* 2026

- Engineered a layered governance system inventorying EC2, EBS, S3, IAM, RDS, CloudWatch, and security groups via eight paginated Boto3 service managers, flagging idle resources against a JSON rule engine.
- Designed an event-driven path where an AWS Lambda handler on an EventBridge cron rule executes cleanup and publishes structured summaries to SNS; a local scheduler reuses the same engine on-premise.
- Made deletions safe by default via dry-run mode, protected-tag filters, timestamped audit logs, and a `unittest` suite mocking AWS APIs; shipped a Flask dashboard over REST endpoints with PDF/CSV exports.

EXAI-ResumeIntel — Resume-to-Role Matching Service | *Python, FastAPI, Pydantic, Docker, Nginx, NumPy* 2026

- Built a similarity-search backend scoring documents against 1M+ job postings by streaming a 4.96 GB memory-mapped float32 embedding array in chunks, bounding resident memory instead of loading it all.
- Replaced an ad-hoc standard-library HTTP server with a typed FastAPI service using Pydantic schemas, and split a monolithic script into six single-responsibility modules behind one coordinator.
- Containerized the stack with Docker, Kubernetes manifests, and an Nginx reverse proxy, offloading 6 GB of artifacts to remote storage behind a lazy auto-downloading loader.

MediXplain — Full-Stack Prediction Web Application | *Python, Flask, REST API, FAISS, pytest, GitHub Actions* 2026

- Developed a full-stack web app on Flask's application-factory and blueprint pattern with validated JSON `/api/predict` and `/api/health` endpoints behind a lazily initialized service singleton.
- Implemented retrieval over a FAISS inner-product index of L2-normalized 384-dim vectors for sub-millisecond top-K lookup, degrading to a TF-IDF retriever when native dependencies are unavailable.
- Configured a GitHub Actions pipeline running `flake8`, `black`, and a 12-test `pytest` suite across a Python 3.10–3.12 matrix, monkeypatching paths so tests run without trained models.

Strider Runner — 2D Game Engine Systems | *C#, Unity, GitHub Actions, Git LFS*

2026

- Architected a 47-file C# codebase across four decoupled layers (input, simulation, presentation, persistence) with a generic singleton base class and data-driven ScriptableObject configuration, and automated Windows/WebGL builds plus edit- and play-mode test runs through a GitHub Actions workflow.

Publications

Automated AWS Resource Cleanup for Optimization of Cost and Security

Proc. 2nd Intl. Conf. on Advanced Nexus of Data and Information Technology (ICANDIT 2026), INTI Intl. Univ., Malaysia, 2026.

Technical Skills

Languages: Python, C++, C, Java, C#, SQL, JavaScript

Backend & Web: REST APIs, Flask, FastAPI, Streamlit, HTML/CSS, application architecture

Cloud & Infrastructure: AWS (EC2, S3, EBS, IAM, Lambda, EventBridge, CloudWatch, SNS), Boto3, Docker

Software Engineering: Linux, Git, GitHub Actions (CI/CD), pytest, unittest, mocking, flake8, black, Makefiles, YAML/JSON

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, System Design

Achievements & Certifications

Outreach Head, TechnoVIT & Vibrance, VIT Chennai (2024–2025) — Led a 25+ member team driving sponsorship, publicity, and media coordination across two flagship university festivals.

First Runner-Up, Figma × Apple Vision Pro Design Challenge, GDSC VIT Chennai (2023) — Designed and prototyped a spatial computing UI using Figma; recognized for creativity and UI/UX innovation.

Certifications: Google UX Design Professional Certificate; Intro to Generative AI (Google Cloud); Python Data Structures & Programming for Everybody (Univ. of Michigan, Coursera).